

Java OOP Assignment (Exercises 10.12 – 10.17)

Exercise 10.12 – Payroll System with Birthday Bonus

```
class Date {
    private int month, day, year;
    public Date(int m, int d, int y){ month=m; day=d; year=y; }
    public int getMonth(){ return month; }
    public String toString(){ return day+"/"+month+"/"+year; }
}

abstract class Employee {
    private String name;
    private Date birthDate;
    public Employee(String n, Date b){ name=n; birthDate=b; }
    public Date getBirthDate(){ return birthDate; }
    public abstract double earnings();
}
```

Exercise 10.13 – Shape Hierarchy

```
abstract class Shape { public abstract String getDescription(); }

abstract class TwoDimensionalShape extends Shape {
    public abstract double getArea();
}

abstract class ThreeDimensionalShape extends Shape {
    public abstract double getArea();
    public abstract double getVolume();
}
```

Exercise 10.14 – PieceWorker

```
class PieceWorker extends Employee {
    private double wage; private int pieces;
    public PieceWorker(String n, Date b, double w, int p){
        super(n,b); wage=w; pieces=p;
    }
    public double earnings(){ return wage * pieces; }
}
```

Exercise 10.15 & 10.16 – Payable Interface

```
interface Payable { double getPaymentAmount(); }

abstract class Employee implements Payable {
    public abstract double earnings();
    public double getPaymentAmount(){ return earnings(); }
}
```

Exercise 10.17 – CarbonFootprint

```
interface CarbonFootprint {
    double getCarbonFootprint();
}

class Car implements CarbonFootprint {
    private double fuel;
```

```
public Car(double f){ fuel=f; }  
public double getCarbonFootprint(){ return fuel * 2.3; }  
}
```